SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Music Streaming Application

1. Introduction

1.1 Purpose

The purpose of the Music Streaming App is to provide users with an easy platform to search, stream, and manage music online. Users can listen to songs, create playlists, like songs, and manage their profiles.

1.2 Scope

- The application allows users to:
- Register and login.
- Search for songs, artists, and albums.
- Stream music online.
- Create and manage playlists.
- Like or unlike songs.
- View recently played songs.
- Control music playback such as play, pause, next, and previous.
- Manage their user profile.

1.3 Users

- User: Searches and listens to music and manages playlists.
- Admin: Manages songs, artists, albums, and users.

2. Functional Requirements

FR1 – User Registration

The system shall allow new users to create an account using details such as name, email, and password.

FR2 – User Login

The system shall allow registered users to log in using valid credentials.

FR3 – Search Music

The user shall be able to search for songs, artists, albums, and playlists.

FR4 – Music Streaming

The system shall allow users to stream and listen to available songs through the internet.

FR5 – Music Controls

The user shall be able to:

- Play
- Pause
- Resume
- Stop
- Skip to next song
- Play previous song
- Adjust volume

FR6 – Playlist Management

Users shall be able to create, edit, and delete playlists and add or remove songs.

FR7 – Like/Favourite Songs

Users shall be able to like songs and view their favourite songs.

FR8 – Recently Played

The system shall maintain a list of songs recently played by the user.

FR9 – Profile Management

Users shall be able to view and update their profile information.

FR10 – Admin Management

The admin shall be able to add, update, and remove songs, albums, artists, and user accounts.

3. Interface Requirements

3.1 GUI Interface

The application shall provide a simple and user-friendly graphical interface.

Main screens include:

- Login/Register Screen

- Home Screen
- Search Screen
- Music Player Screen
- Playlist Screen
- Favourite Songs Screen
- User Profile Screen
- Admin Dashboard

The music player shall contain buttons such as Play, Pause, Next, Previous, Volume, and Progress Bar.

3.2 Hardware Interface

The application may interact with the following hardware:

- Smartphone/Computer
- Speakers or headphones
- Touchscreen or keyboard and mouse
- Internet/network hardware
- Storage devices for temporary music data

The device should have sufficient processing power and memory to run the application smoothly.

3.3 Software Interface

The application may interact with:

- Operating System: Android, Windows, Linux, or iOS
- Database: MySQL / MongoDB
- Server: Application/Web server
- Internet: Required for online music streaming
- Authentication service: For secure user login
- Media player: For playing audio files

4. Non-Functional Requirements

4.1 Security

- User passwords shall be stored securely.
- Only authenticated users can access their accounts.
- User information shall be protected from unauthorized access.
- Admin functions shall be accessible only to authorized administrators.

- The application should use secure communication such as HTTPS.

4.2 Reliability

- The application should play music without frequent interruptions.
- User playlists and favourites should not be lost.
- The system should handle errors without crashing.
- Database information should be backed up regularly.
- The system should recover properly after temporary failures.

4.3 Availability

- The application should be available 24/7, except during scheduled maintenance.
- Users should be able to access their account whenever the service is available.
- The server should minimize downtime.
- The system should handle multiple users simultaneously.

Conclusion

The Music Streaming App provides users with a convenient platform to search, stream, organize, and manage music. The SRS defines the functional, interface, security, reliability, and availability requirements needed to develop the system.